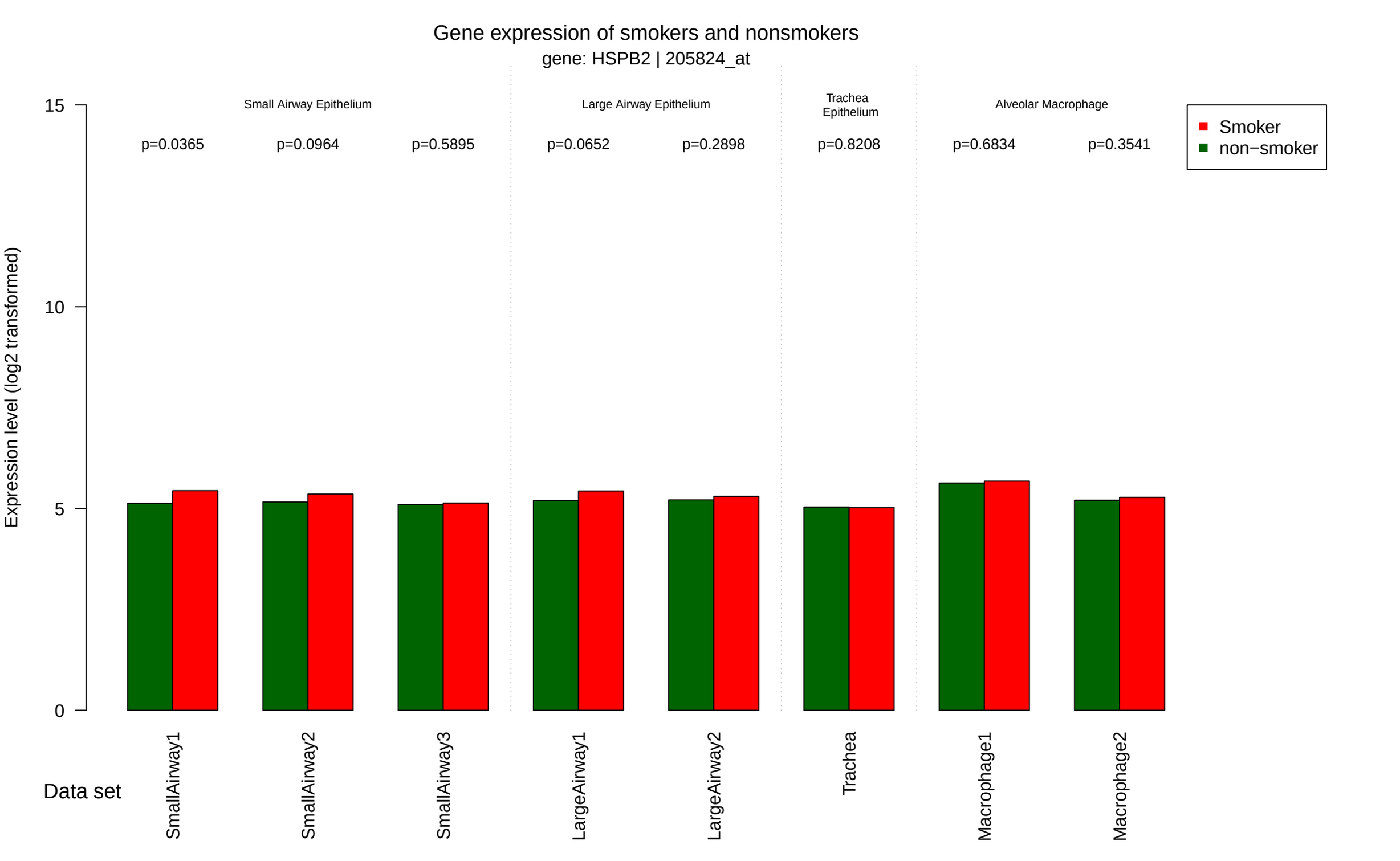
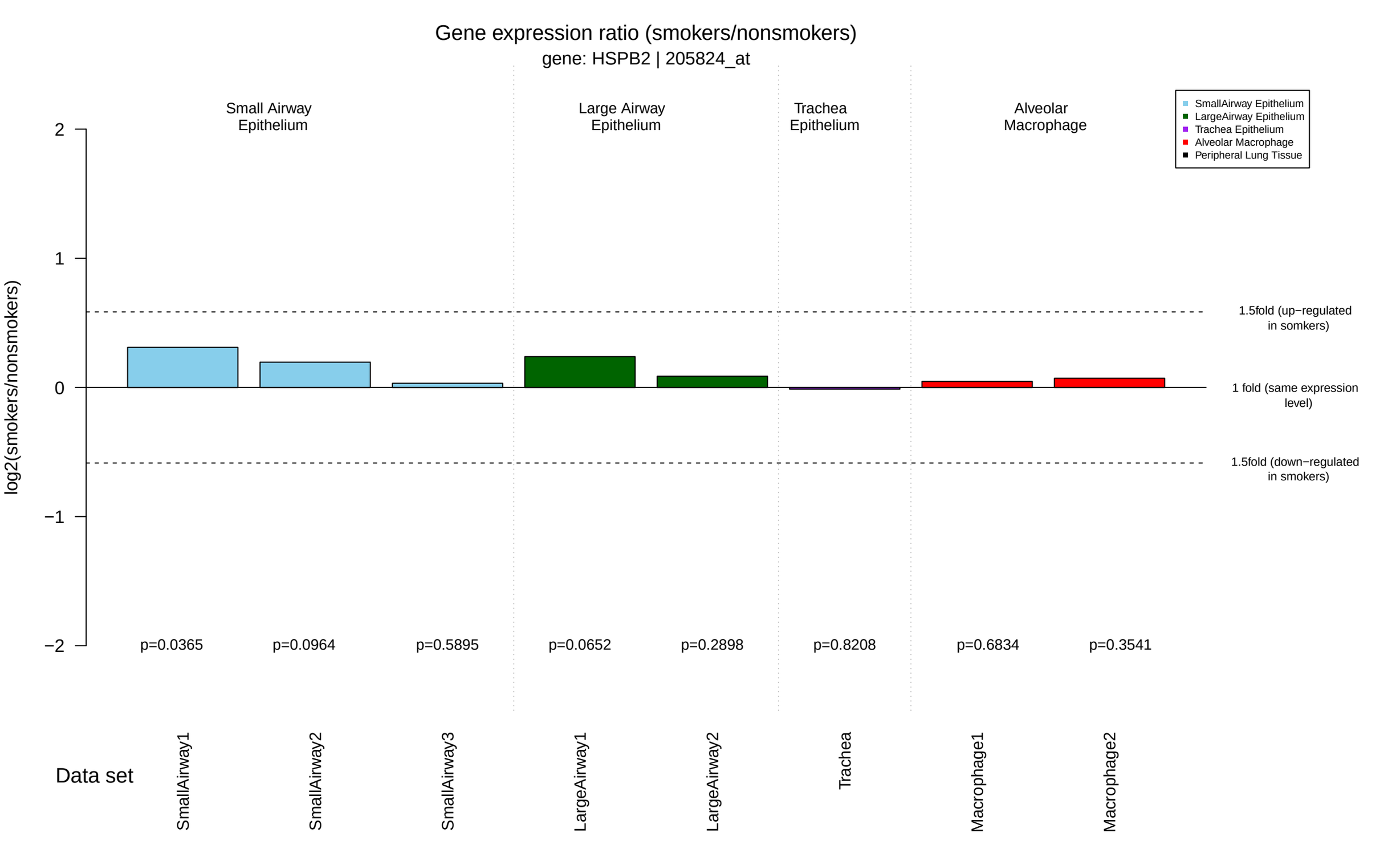
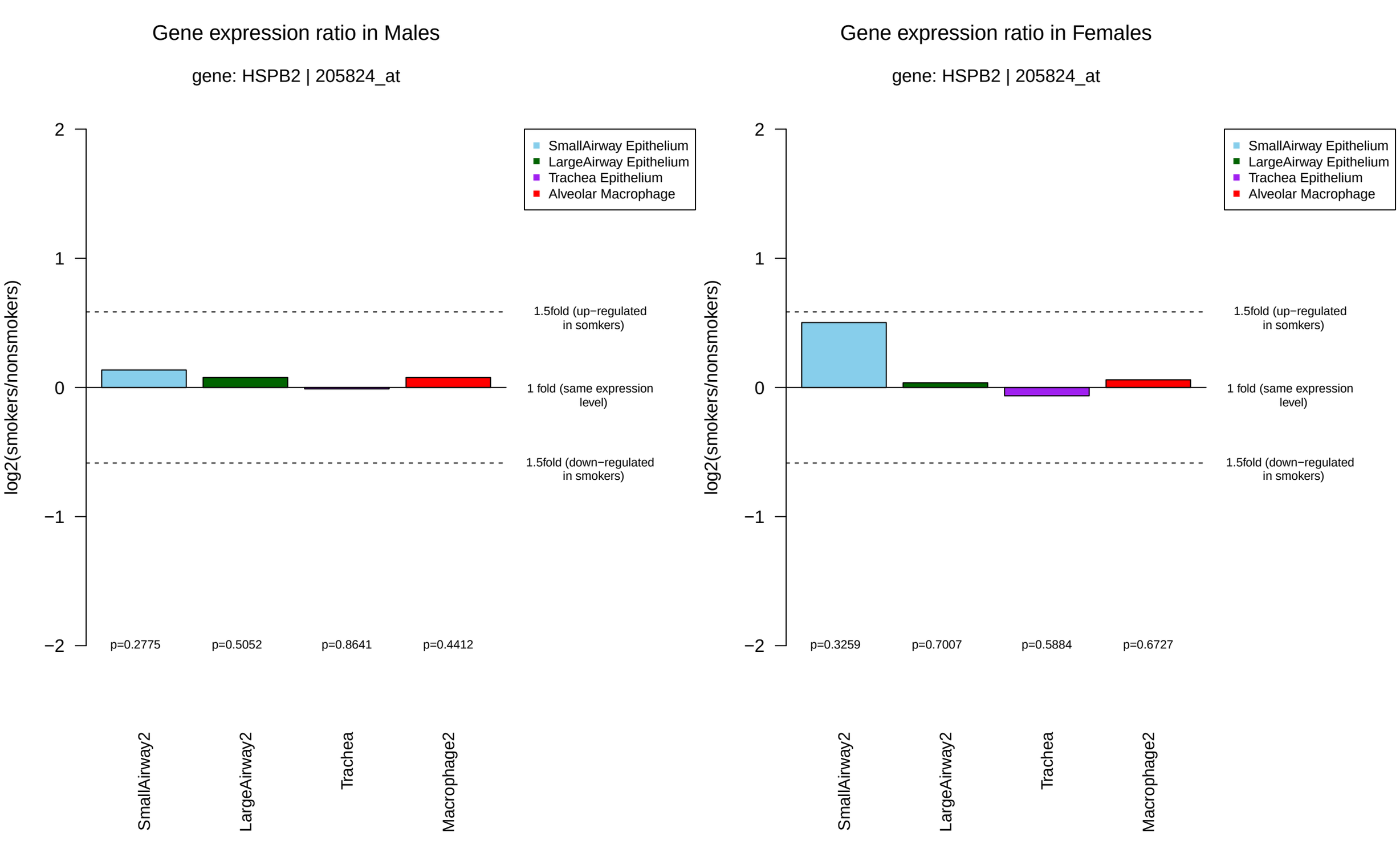
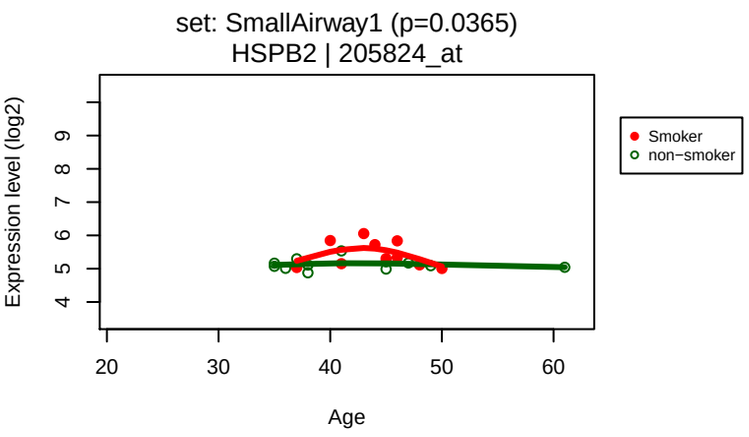
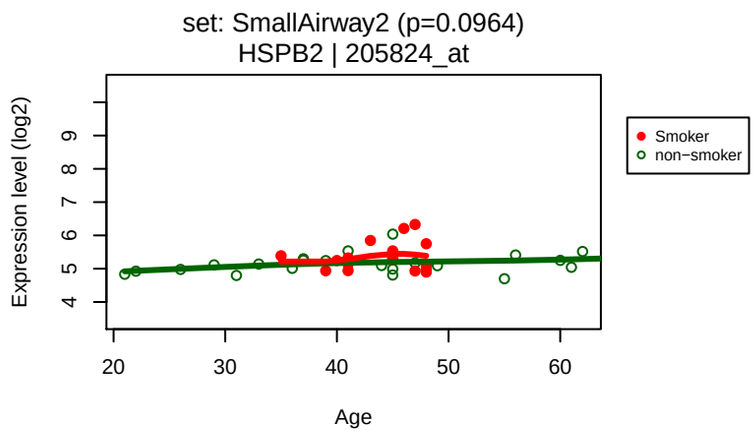
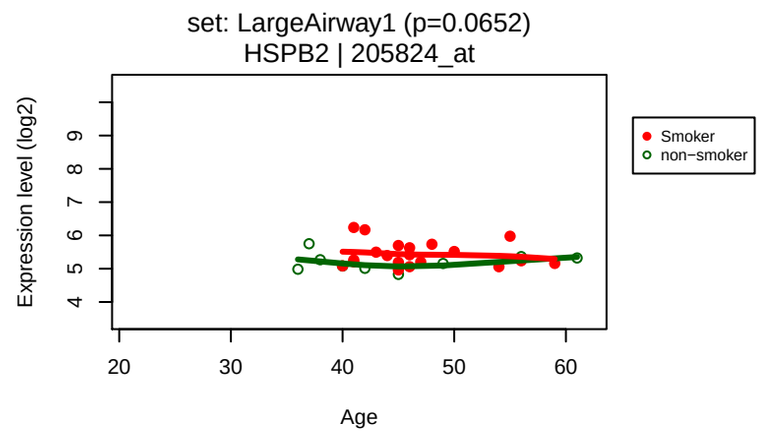

Gene expression of smokers and nonsmokers
gene: HSPB2 | 205824_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0365
p=0.0964
p=0.5895
p=0.0652
p=0.2898
p=0.8208
p=0.6834
p=0.3541
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

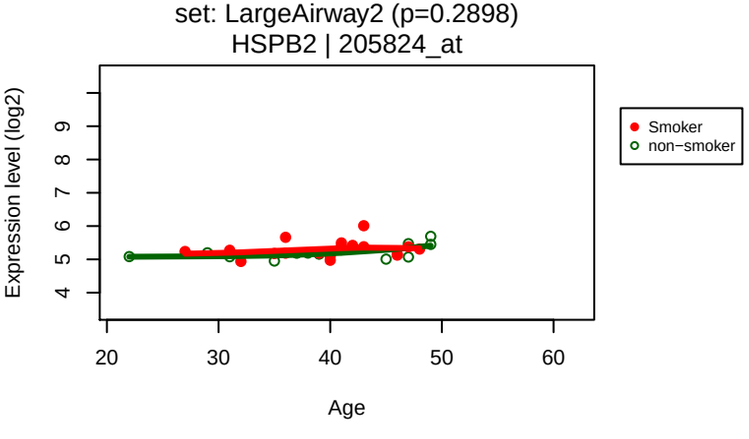

Gene expression ratio (smokers/nonsmokers)
gene: HSPB2 | 205824_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0365
p=0.0964
p=0.5895
p=0.0652
p=0.2898
p=0.8208
p=0.6834
p=0.3541
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

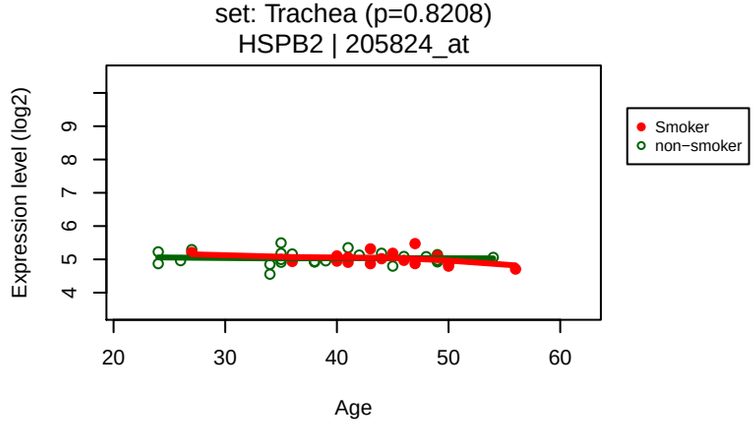

Gene expression ratio in Males
gene: HSPB2 | 205824_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2775
p=0.5052
p=0.8641
p=0.4412
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPB2 | 205824_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3259
p=0.7007
p=0.5884
p=0.6727
SmallAirway2
LargeAirway2
Trachea
Macrophage2

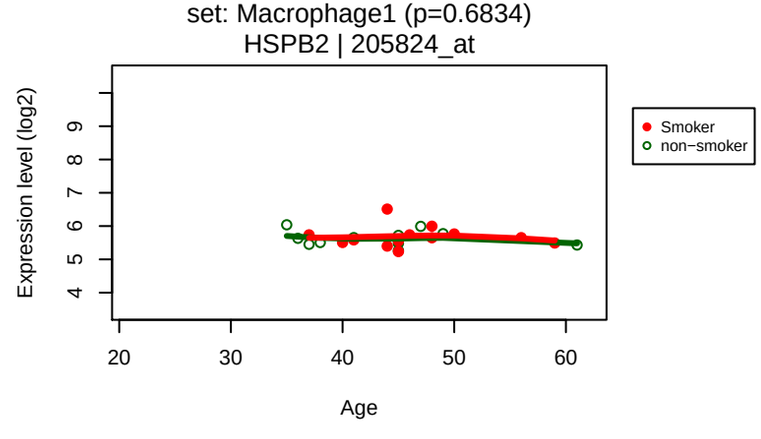

set: SmallAirway1 (p=0.0365)
HSPB2 | 205824_at
Expression level (log2)
Age
Smoker
non−smoker

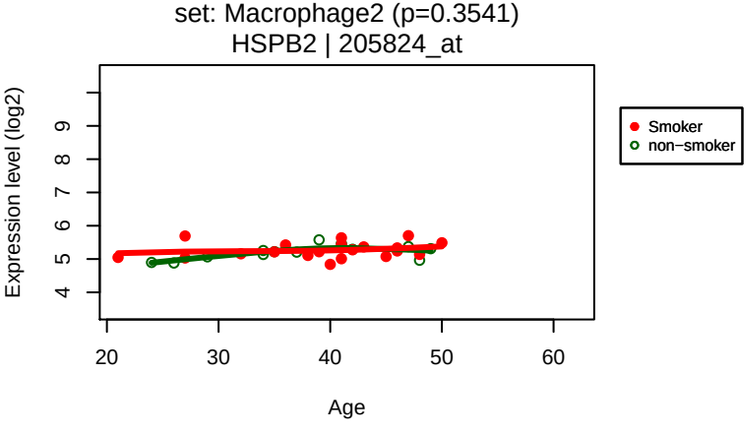

set: SmallAirway2 (p=0.0964)
HSPB2 | 205824_at
Expression level (log2)
Age
Smoker
non−smoker

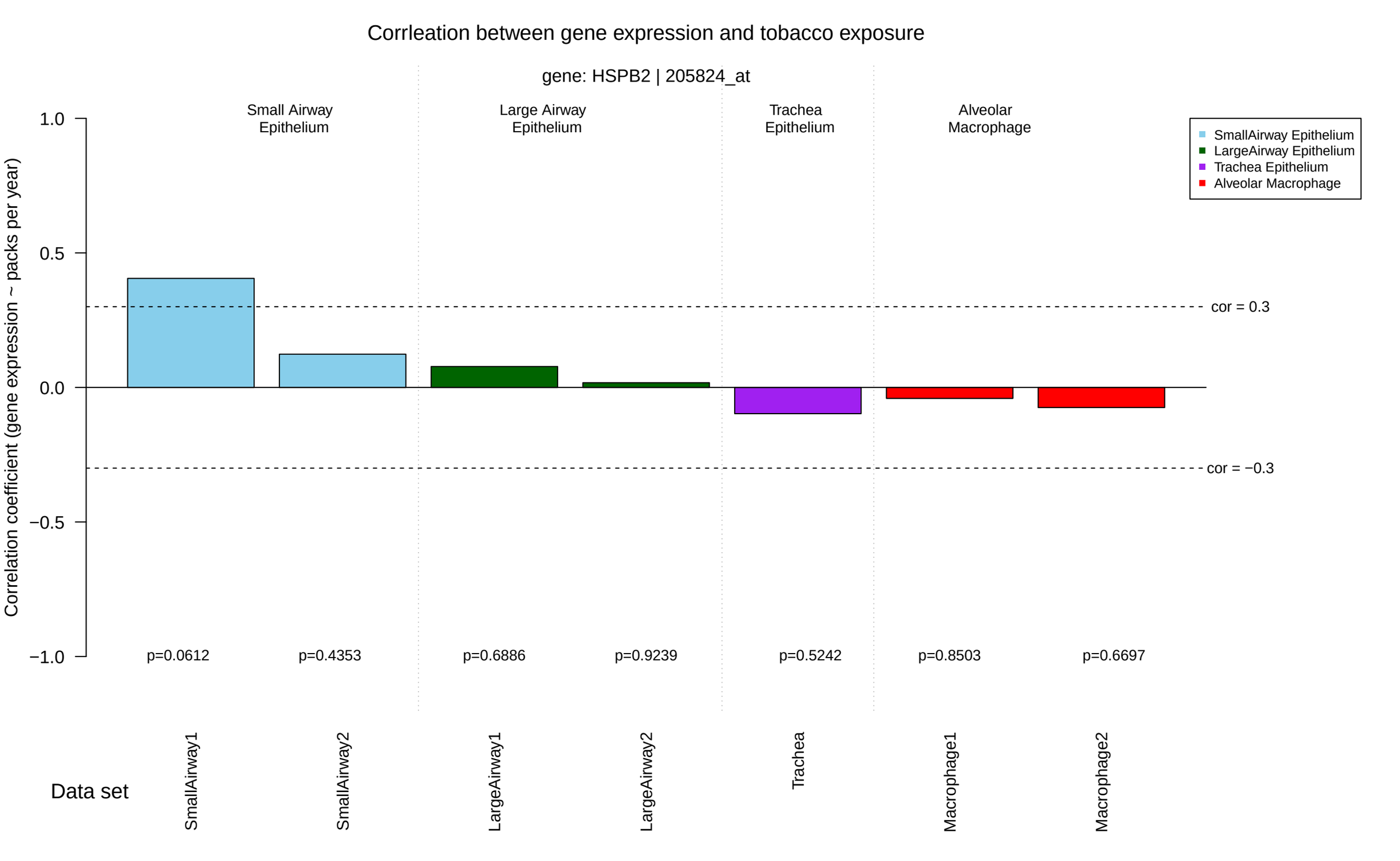

set: LargeAirway1 (p=0.0652)
HSPB2 | 205824_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2898)
HSPB2 | 205824_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8208)
HSPB2 | 205824_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6834)
HSPB2 | 205824_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3541)
HSPB2 | 205824_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPB2 | 205824_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0612
p=0.4353
p=0.6886
p=0.9239
p=0.5242
p=0.8503
p=0.6697
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2